

Estes Park Group

Preliminary themes and observations from the data-centric retreat

June 2017 kickoff

Dave McComb organized and conducted the brainstorming sessions at this retreat, which took place June 8-11 in Estes Park, Colorado. At the tail end of the retreat, attendees chose the name Estes Park Group for themselves.

The first goals of the Group are to articulate a vision for a SMART* (see **SMART* Meaning map** theme: Best practices in semantic data architecture) semantically enabled, model-driven, data-centric, application development and to propose suitable reference architecture that would support it.

The kickoff gathering discussed primary themes around the problems the movement hopes to address and the solutions the movement proposes.

Participants:

- Mike Bowers
- Tim Cook
- Ted Hills
- Randy Kahle
- Dave McComb (organizer and host)
- Parsa Mirhaji
- Alan Morrison
- Tim Smith

Below is a straw man draft of the themes discussed, a starting point for capturing the main takeaways from the retreat. Attendees, please feel free to edit and add to this document as appropriate.

Ted proposed an appropriate tagline for the Group: Durable meaning, lasting value. He also suggested the Group identify tools that would be useful in implementations of the architecture.

In the straw man it isn't possible to expand upon everything that was discussed, but when others add themes not listed in a revision, we can make this document more comprehensive. {Randy, I missed part of your talk and wasn't able to contextualize and place it here. Would you be able to add a theme or two?}

Data-centric theme: What is "data-centric" from our perspective?

Tim C. did some help thinking that the Group appreciated on what data-centric should be from our perspective. He identified eight characteristics that can be associated with true data-centricity:

1. **Importance.** Data is the prime asset of any organization.
2. **Permanence and preserved full fidelity.** Data migrations are costly, both in the process and in information loss. It should be possible to store data for an unlimited amount of time.
3. **Contextually derived information.** Information is data with context.
4. **Logical, temporal and spatial semantics.** Context is the combination of ontological (what), temporal (when) and spatial (where) and other semantics about the data.
5. **Semantically-derived knowledge.** Knowledge is derived from information managed over time.
6. **Model flexibility and openness.** An information provider today cannot know the use cases of information consumers of tomorrow. Therefore, creating models upfront with the complete context that will fit all the use cases forever is impossible.
7. **Managed for shareability and auditability.** Data models and information instances must be computable, sharable, immutable, traceable and uniquely identifiable.
8. **Preserved and immutable.** Proper information modeling must be future-proof; and immune to semantic shift/drift. no data is ever left behind.

Data access theme: Concept of “data-centric” as others view it and how it could prove useful in data de-siloing/accessibility improvement efforts

Group consensus was that data without meaning is valueless. Most concepts that describe “data-centric” don’t consider how to address the problem of scant, hidden, ambiguous, or missing meaning in data.

At the same time, it would be best to leverage the term “data-centric” as others view it and contrast it with “application-centric”. The application-centric mentality continues to be unfocused and wasteful. It perpetuates the mistakes of previous generations of enterprise information tech, something Dave has been/will be writing about.

The term “data-centric,” Tim S. pointed out, does emphasize the need to emancipate and unleash the data for reuse first and foremost. Unless organizations can eliminate data cartels and take a scalable model-driven approach, they can’t make more effective reuse of large amounts of data being generated.

Most corporate stories trumpeted in the press on the effective and novel use of advanced analytics that ends up in the press describe the narrow impact of a tactical project. Often a data science team is tapped or brought in from an outside consultancy and has to struggle just to get data to work with. In other cases, teams with operational responsibilities trying to build operational efficiencies into the information flow are thwarted by “data cartels,” a term a Coca-Cola executive coined.

The data the team manages to get (which in the worst-case scenario may be just data that originates in the public sector or bought from a third party) must be thoroughly cleaned and prepared for the project, consuming the majority of the time of the team. The team uses the tactically prepared data to model and generate decision-making recommendations. The quality of this data is questionable because the original meaning and organizational context is lost during the data massaging effort. Considerable time is then spent compiling and creating one-off visuals for a project report, or series of reports.

This approach clearly is appropriate only for pilots and proofs of concept and has zero scale economies. Reading between the lines of the stories of these projects, it becomes evident these companies have no effective, integrated data foundation across application silos. If they did, the impact of even garden-variety analytics would be substantial.

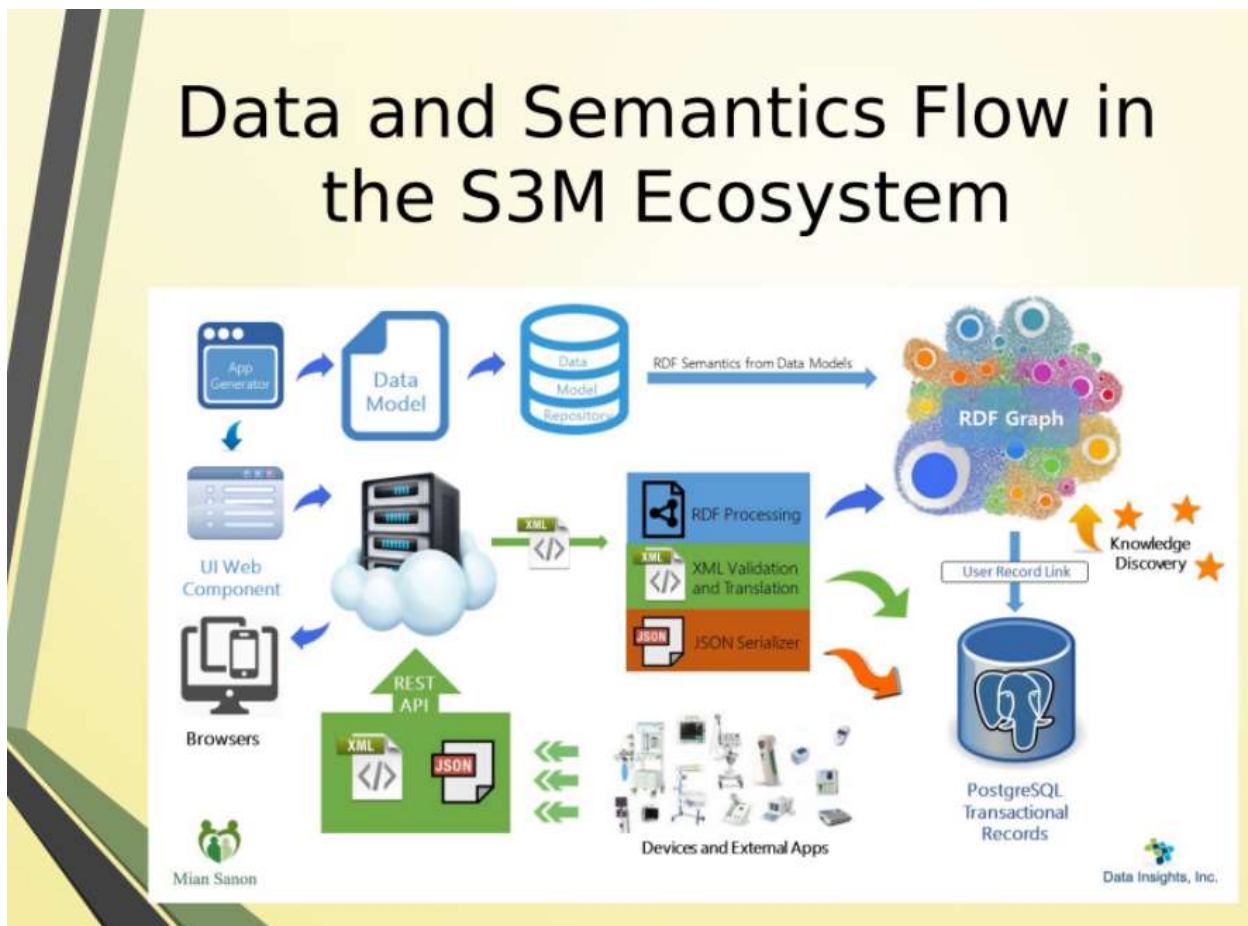
The lack of an effective data foundation in the highly competitive and volatile present-day marketplace is the result of a myopic, application-first mindset that keeps perpetuating itself with each new generation of information technology companies adopt. Most recently, companies that have been moving to cloud architectures often just recreated the application functionality and data silos they had, and then business units have added to the data sprawl with new SaaS applications. To reuse the data in those applications, companies are forced to start with only semi-structured CSV data.

Shareable, Structured, Semantic Model (S3Model) and shared meaning theme.

Tim C. pointed out that the S3Model he led the development of beginning in 2009 once validated provides a systems concept the Group could adapt and adopt. Core to the model is the notion of shared meaning.

Shared meaning is created by using a model-first approach where models are derived by constraint, from a small broad definition Reference Information Architecture. Immutable, executable, standards-based models capturing the complete context of intent at creation time are shareable across applications inside and outside of organizations as well as across domains of knowledge. All data is then related to these semantic models.

Data and Semantics Flow in the S3M Ecosystem



ECO-SYSTEM DIAGRAM

SMART* Meaning map theme: Best practices in semantic data architecture
 {Mike—you can articulate this theme better than I can.}

Mike has conceptualized an architecture and could build a prototype system directed by the reference architecture the Group adopts.

Within the context of this movement, Mike coined an acronym, more or less as follows:

Semantically Mapped Atomistic Represented Transfer

Mike observed that graph structures are atomistic (a term borrowed from the definition of ACID).

Best practice, Mike points out, is a flow with both an input and output semantic mapping, which could be called a “shared meaning map”. Before data flows into the lake (or hub), meaning must be preserved and contextualized as follows:

Meaning map data treatment process for lake data

Data ingestion → data and meaning map of input to preserve → map to target graph → context selection for output → output for reuse

Consider how this approach differs from that most commonly used for data lakes, which merely addressed the accessibility problem. The rationale behind data lakes was aptly described by Mike Lang of Revelytix (now part of Teradata) before he retired in 2014:

“C-level executives are saying, “It’s no longer inordinately expensive for us to store all of our data. I want all of you to make copies. OK, your systems are busy. Find the time, get an extract, and dump it in Hadoop.”

But they haven’t integrated anything. They’re just getting an extract.

Data de-siloing and graph structuring at scale, mindful data generation and collection for purpose and broad integration and interoperation would magnify the impact of an analytics capability to the point that it can clearly be transformative for the organization. This strategic approach often eludes most enterprises, for reasons beyond the scope of this document.

Model-driven theme: *Flexible meaning that’s machine-readable encoded as data that lives with the data—the key to data management automation at scale*

Meaning that’s made explicit and machine-readable is what enables integration and reuse.

Machine readable meaning that lives as data alongside the data it contextualizes provides the key to data management automation. That level of automation in data-centric systems is essential for advanced analytics and machine learning at scale. Otherwise, advanced analytics is a one-off that’s separately managed, with no economies of scale possible.

The trade press is full of stories about the benefits of machine learning, but when reading between the lines, it’s apparent those success stories have to date been local to individual projects. What semantic graph-based data lakes provide is persistent, broadly available contextualized information that users can tap for **a range of** projects simultaneously, not just one project at a time, as the Montefiore example described below underscores.

For process improvement, organizations need to build data-centric learning loops. The machine learning paradigm implies construction of these loops. With machine-readable meaning woven into the data in the semantic data lake or hub, tapping into numerous new data sources and harnessing them for reuse can be made automatic. Reasoning and inferencing boost the richness and integration potential of aggregated sources:

Data → reasoning → data

With a semantic graph model-driven approach, contextually relevant knowledge, recommendations, and prescriptive decision support can be put within reach of the decision maker. Machine readability/interpretability makes this approach scalable.

Semantic graph as master data model theme

The various families of NoSQL databases as Mike described them can be arranged in terms of data model complexity, from key-value to semantic graph. Document trees, for example, are extensible but simpler than graphs. Tables are less extensible but can also be represented in graphs.

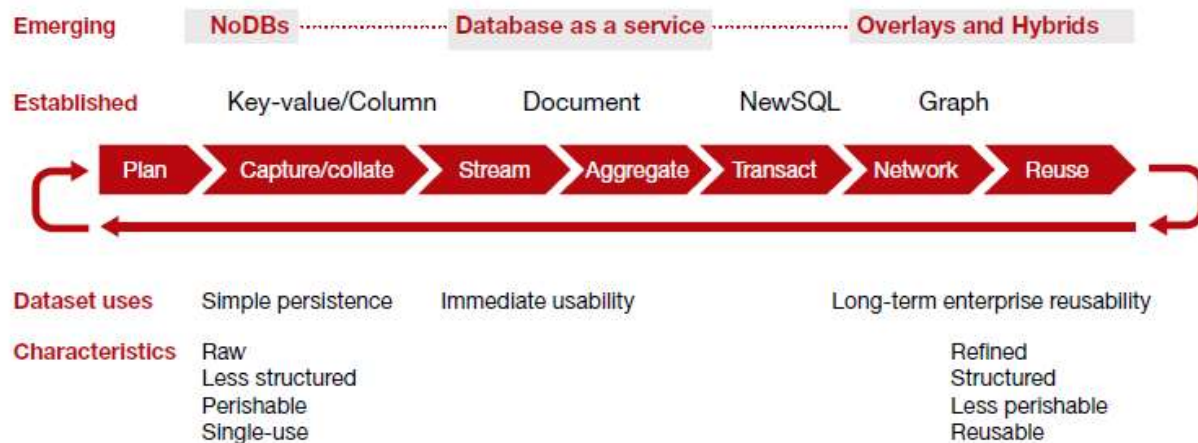
Key value—Wide column—Document—newSQL/relational—tactical graph—semantic graph

The latter becomes the parent and the other databases children. In that sense, the architecture accommodates all database families and yet makes it possible to offer a master data lake or inclusive data warehouse, with connectors to database model X, Y or Z. Semantic graphs as the most expressive family can express and contain all the other families.

The more expressive and structured the data model, the slower the database is. Tactical, one-off data can be handled by databases with stripped-down data models such as key-value or wide column. Strategic reuse and permanent data are at the far end of this continuum in this illustration Alan developed for PwC in 2015.

The nature of data-centric paradigm, the need to preserve good data for analytics purposes, and the advance of in-memory technology are all trends that favor semantic graphs.

NoSQL trends and the data lifecycle



Low-code/no-code theme: Data (that's fully explicit and expressive) as code

The Estes Park Group told several stories about the scale economies of the model-driven approach. What's distinctive about the semantic graph approach to model-driven development as Ted pointed out is that the data once triplified with the help of an appropriate ontology or semantic graph model **becomes** the code. It's machine-readable data that can function as code with the help of an interpreter.

In this sense, the humans build the model up front, a data-centric, no-code method. The interpreter does include some code. The model-driven "code" that's part and parcel of the triplified data is dynamic and flexible, not brittle and static, declarative rather than procedural. In that sense, it's designed to scale.

Those who start with the data and structure it in this way tend to extend the notion of a query language along the lines that Holger Knublauch of TopQuadrant, for example, has done with SPARQL Motion. In essence, as Tim S. pointed out, this extended query language suggests an additional "application" paradigm.

Load and learn theme. Parsa first pointed out that his Montefiore architecture dispenses with the transform step. Then after some discussion of architecture we decided it dispensed with the extract step too. Analytics-ready data in the *Semantic model-driven, data-centric case study* below implies that SMART data can just be loaded into the learning loop.

Relations versus relationships, type versus class and other distinctions theme. Ted has written extensively on ambiguous language and how developers who aren't data savvy and data architects, engineers, etc. are always talking past each other. He authored a book called [NoSQL and SQL Data Modeling: Bringing Together Data, Semantics, and Software](#), blogs at <http://tdan.com/author/ted-hills> and has written a programming language that among other things seems to address sources of ambiguity and clear a path for semantically aware, model-driven, data-centric development.

Human- versus machine-scale theme: Human-scale views, but machine-scale manageability

Humans need simple views of the data or abstracted visualizations. They cannot comprehend tables with 1,000 columns. Tables with 1,000 columns, Dave pointed out, might as well just be unstructured data.

Human limits of understanding are even more apparent when it comes to graphs. For that reason, It's good to present humans with simple, human comprehensible models.

Machines, for their part, can handle n-dimensional graphs. The challenge therefore becomes how to push all the complexity machines can handle to machines, and tailor what ends up at the presentation layer or user interface so that's it's human comprehensible.

Collapsing the stack theme

Keep in mind that legacy application-centric architectures have lots of artificial constructs and layers that can now be dispensed with in a semantic graph, model-driven, data-centric approach. EnterpriseWeb, a data and logic virtualization + process integration platform vendor, calls the elimination of these constructs and layers “collapsing the stack”. A before-and-after architecture illustration could depict this kind of simplification.

Path to machine intelligence through data quality theme

Would make sense to hitch the SMART model-driven, data-centric architecture to the growth of machine learning. In essence, AI’s prominence in the minds of CxOs makes it necessary to ensure close connection between a reference architecture and AI. Tim C. pointed out that with the SMART data architecture, we’re solving the data quality problem that hampers the effectiveness of machine learning.

How do companies get started? theme

The Group discussed lots of different challenges, lessons learned when it comes to pitch or guerrilla adoption strategies. It was evident that Parsa has leadership buy in and support that allowed him to continually evolve the system he’d envisioned.

Tim S. came up with these helpful mnemonic taglines:

Think big; start small.

Think strategically; act tactically. Show something.

Semantic model-driven, data-centric case study:

Montefiore Health System/Einstein College of Medicine

{Parsa—there may be technical or other inaccuracies or gaps in this quick writeup; please check and correct or augment as appropriate.}

Parsa Mirhaji and team at Montefiore/Einstein have built a model-driven, data-centric system that enables an integrated, real-time, analytics-ready feed from a continually updated semantic data lake that’s now part of Montefiore clinical evidence generation framework.

The source data is a mix of data types and sources, some of which are real-time, others batch (such as from a knowledge graph that includes structured Snomed CT, etc. nomenclature, comprehensive medical reference content etc.). The target data lake is a semantic integration graph that has connectors to the EMR DB and other data modeled in other ways.

The graph allows output model-driven, contextualized feeds from the integrated lake data via a simple, domain-specific, patient-centric ontology and an interpreter. The feed can be accessed via the Frame Description Language (inspired by R data frames) provided or via other methods. You could imagine an option for a chatbot front end with NLP enriched and contextualized by the structured data in the semantic data lake.

In some ways it’s a more transparent and functional Watson, in others it’s patient-centric data management automation, a Cray Urika like approach for a variety of analytics purposes. Machine learning wise, Montefiore has been using random forest and Recurrent Neural Network (RNN) algorithms to date for the machine learning part of the analytics platform, but you could imagine a platform that offers a fully heterogeneous library. Results of ML model training are also fed back into the lake for enrichment purposes. Data scientists can use an interface that Parsa calls an FDL (Frame Description Language) to select a context of interest.

Graph visualization is via Gruff, Franz’s graph visualization tool, but the environment also has connectors via the FDL frame to KNIME, SAS, H2O.ai, or other tabular data consuming analytics suites, and users could tap the visualization functionality of those systems. Franz Allegrograph is the distributed semantic graph database used for the semantic data lake.

Currently the system is running in parallel to conventional EDW core and BI systems, and it’s designed to analyze health evidence continually collected from 66,000 inpatient encounters annually, and provide real-time, patient-relevant decision support to the hospital’s physicians.

Montefiore is a non-profit chain of 11 hospitals in NY and CT focused on driving down costs. Once they scale up past a certain point (1.3m capitated patients), the theory goes, profitability will improve. This is a strategy (absent the price reductions) seemingly comparable to what Jeff Bezos espoused for Amazon years ago, according to Tim O'Reilly:

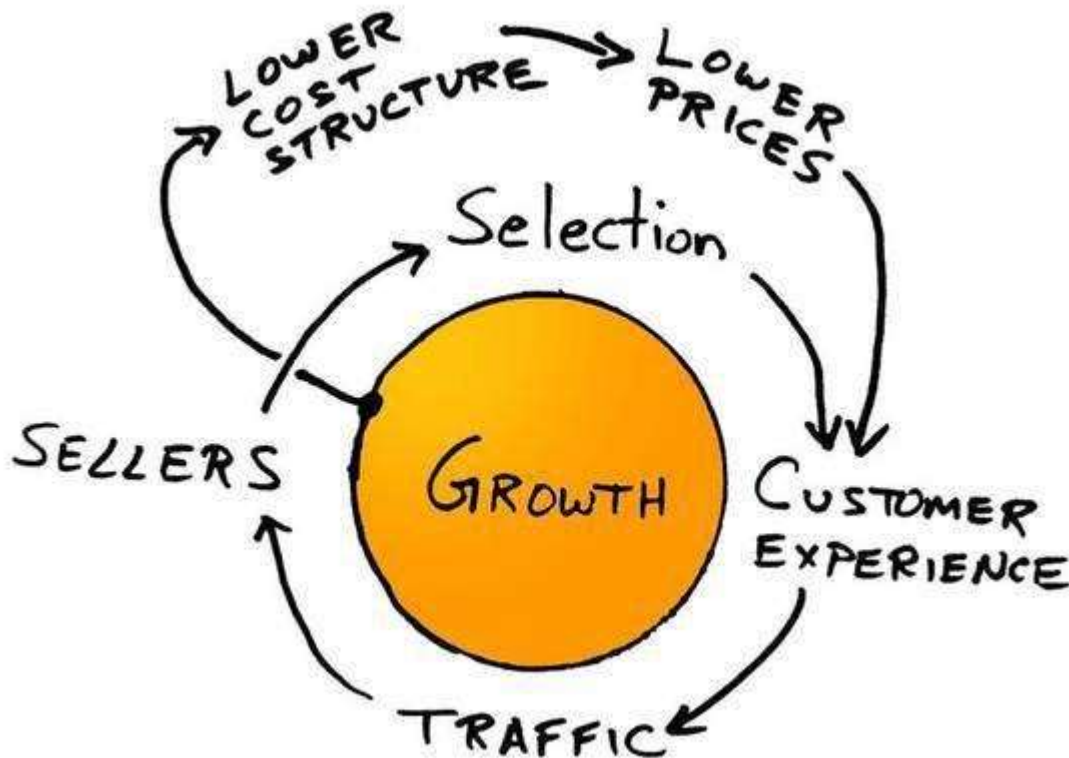


Image via Ben Evans, ["Why Amazon Has No Profits, and Why It Works."](#) 2014.

Parsa's developed his methods over decades as a cardiologist and data architect first at the University of Tehran and then at University of Texas before being hired at Montefiore. Funders here in the US over the years have included TATRC, the National Heart, Lung, and Blood Institute (NHLBI)/NIH, NSF. Intel and Cisco have donated equipment, and for Intel's part they've donated Optane advanced, persistent, non-volatile DDR RAM for low-latency big data storage.

Under earlier federal contracts and grants through TATRC, he's developed bio surveillance systems that were in use after 9/11 and during Hurricane Katrina and Rita.